

Supplemental Material for “Rayleigh Bound State in the Continuum from Hybridized Acoustic Cavities”

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CONVENTIONS AND FINAL GEOMETRY

The structure is invariant along z , periodic along x , and occupies the half-space $y \leq 0$. Water fills $y > 0$ and the two grooves; every solid boundary is treated as acoustically rigid. The convention is $p \propto e^{+i\omega t}$. Outgoing propagating waves vary as $e^{-ik_y y}$, while evanescent waves have $\text{Im } k_y < 0$.

The final design was obtained at the exact $n = -1$ Rayleigh relation $\Omega = 1 - \kappa$. Its normalized and dimensional parameters are listed in Table S1. The dimensional realization uses $c = 1500 \text{ m s}^{-1}$ and $f_{\text{BIC}} = 180 \text{ kHz}$.

The Bloch point is

$$\begin{aligned}\kappa_{\text{BIC}} &= 0.110000105311808, \\ \Omega_{\text{BIC}} &= 0.889999894688192, \\ \theta_{\text{BIC}} &= 7.0996629128^\circ.\end{aligned}\quad (\text{S1})$$

At this point $n = 0$ is propagating, $n = -1$ is exactly grazing, and $n = +1$ is evanescent. Negative κ gives the reciprocal partner and exchanges $n = -1$ with $n = +1$.

FULL MODAL-MATCHING OPERATOR

The exterior field of a homogeneous mode is

$$p^{\text{hom}}(x, y) = \sum_{n=-M}^M A_n e^{-ik_{x,n}x - ik_{y,n}y}, \quad k_{x,n} = k_x + \frac{2\pi n}{a}, \quad (\text{S2})$$

with $N = 2M + 1$ retained orders. In groove ℓ ,

$$p_\ell(x, y) = \sum_{q=0}^{K-1} C_{\ell q} \cos\left[\frac{q\pi(x - x_\ell)}{w_\ell}\right] \cos[\beta_{\ell q}(y + d_\ell)], \quad (\text{S3})$$

TABLE S1. Final two-groove Rayleigh-BIC geometry.

quantity	normalized	dimensional (mm)
period a	1	7.416665789
large-groove depth d_1	0.6941596665	5.148350251
small-groove depth d_2	0.1785417985	1.324184849
large-groove width w_1	0.5944557161	4.408879373
small-groove width w_2	0.1352646233	1.003212504
intergroove gap g	0.1351398303	1.002286956

where

$$\beta_{\ell q}^2 = k_0^2 - \left(\frac{q\pi}{w_\ell}\right)^2. \quad (\text{S4})$$

No single-longitudinal-mode approximation is made. In particular, the large groove retains the transverse $q = 1$ family that dominates the final state.

Define the aperture projection

$$\Pi_{n,\ell q} = \frac{1}{w_\ell} \int_{x_\ell}^{x_\ell + w_\ell} e^{-ik_{x,n}x} \cos\left[\frac{q\pi(x - x_\ell)}{w_\ell}\right] dx. \quad (\text{S5})$$

Projection of pressure and normal-velocity continuity gives

$$\mathbf{F} \begin{bmatrix} \mathbf{A} \\ \mathbf{C} \end{bmatrix} = 0, \quad \mathbf{F} = \begin{bmatrix} \mathbf{B} & -\mathbf{C}_d \\ \text{diag}(k_{y,n}/k_0) & i\mathbf{D}\mathbf{V}_d \end{bmatrix}, \quad (\text{S6})$$

where

$$\mathbf{C}_d = \text{diag}[\cos(\beta_{\ell q} d_\ell)], \quad (\text{S7})$$

$$\mathbf{V}_d = \text{diag}[(\beta_{\ell q}/k_0) \sin(\beta_{\ell q} d_\ell)]. \quad (\text{S8})$$

The groove coefficients are retained as unknowns, avoiding the artificial tangent poles that arise when cavity admittances are formed before the homogeneous eigenproblem.

For driven scattering, the incident and reflected amplitudes are separated from the homogeneous vector. With

$$\mathbf{S} = \mathbf{D} \text{diag}[-i\beta_{\ell q} \tan(\beta_{\ell q} d_\ell)], \quad (\text{S9})$$

the reflected coefficients obey

$$[\text{diag}(k_{y,n}) - \mathbf{S}\mathbf{B}]\mathbf{A}^{\text{dr}} = k_{y,0}\mathbf{e}_0 + \mathbf{S}\mathbf{b}_{\text{inc}}. \quad (\text{S10})$$

For every propagating order,

$$\eta_n = \frac{\text{Re } k_{y,n}}{k_{y,0}} |A_n^{\text{dr}}|^2, \quad \sum_{n \in \text{prop}} \eta_n = 1 \quad (\text{S11})$$

in the lossless calculation.

STRICT RAYLEIGH CRITERION AND ENDPOINT TEST

For $k_{y,n} = 0$, a homogeneous order contributes

$$\int_0^H |p_n|^2 dy = H |A_n|^2. \quad (\text{S12})$$

It therefore remains non-square-integrable even though its normal flux vanishes. At positive κ_{BIC} , strict localization requires

$$A_{-1} = A_0 = 0. \quad (\text{S13})$$

The constrained calculation removes these two amplitudes from the unknown vector and evaluates the scaled reduced operator. For $(N, K) = (345, 43)$,

$$\begin{aligned} \frac{\sigma_{\min}}{\sigma_{\max}} &= 4.9107721 \times 10^{-16}, \\ r_{\text{raw}} &= 9.8731869 \times 10^{-14}. \end{aligned} \quad (\text{S14})$$

This constraint is not used as the only test. We independently evaluate the complete square operator at Eq. (S1), inserting $k_{y,-1} = 0$ exactly rather than relying on a floating threshold classifier. The full operator gives

$$\begin{aligned} \frac{\sigma_{\min}^{\text{full}}}{\sigma_{\max}^{\text{full}}} &= 5.8469558 \times 10^{-16}, \\ r_{\text{raw}}^{\text{full}} &= 1.2808319 \times 10^{-13}. \end{aligned} \quad (\text{S15})$$

The unconstrained endpoint null has $|A_0|/\|\mathbf{C}\| = 2.58 \times 10^{-16}$ and $|A_{-1}|/\|\mathbf{C}\| = 8.09 \times 10^{-17}$. These are the physically relevant amplitude zeros; zero threshold power alone is not used as evidence.

OUTGOING-POLE SPECTRUM AND CONTINUATION

Complex eigenfrequencies were obtained with all-channel outgoing/decaying boundary conditions. Background poles were independently discovered at each sampled positive Bloch wave number. Local line segments were retained only after a global one-to-one assignment passed three hard tests: identical open/closed channel signature, groove-mode overlap exceeding 0.95, and $|\Delta\Omega| < 0.035$. The accepted segments have minimum overlap 0.99875. They terminate rather than being forced across a Rayleigh square-root cut; adjacent-sheet continuations are omitted. The target branch was continued separately with consecutive-eigenvector overlap and has minimum stored overlap 0.99288 over $0 \leq \kappa \leq 0.5$. Guided states are shown in grey and excluded from the radiative- Q color mapping. The color scale spans $0 \leq \log_{10} Q \leq 12$, with high- Q states assigned the most vivid colors. No interpolation or smoothing was used.

The target branch also reaches a symmetry-protected BIC at $\kappa = 0$:

$$\begin{aligned} \Omega_{\Gamma} &= 0.883968664686884 + 2.27 \times 10^{-17}i, \\ Q_{\Gamma} &= 1.95 \times 10^{16}. \end{aligned} \quad (\text{S16})$$

At the same truncation as the Rayleigh endpoint, $\sigma_{\min}/\sigma_{\max} = 9.12 \times 10^{-14}$, $r_{\text{raw}} = 1.83 \times 10^{-12}$, and

$|A_0|/\|\mathbf{C}\| = 1.61 \times 10^{-16}$. The $n = \pm 1$ components are nonzero but evanescent. Thus the Γ state is a conventional symmetry BIC on the same hybridized band, not a second Rayleigh BIC.

Table S2 lists representative points of the target branch. The linewidth is $\Gamma = 2|\text{Im } f_p|$. At the endpoint, the real part of the pole equals the Rayleigh frequency, $\Gamma = 0$, and Q is infinite. The tabulated endpoint amplitudes come from the homogeneous mode, not from a driven reflection minimum.

An independent truncation comparison was performed at $\kappa = 0, 0.11, 0.25$, and 0.5 for the resolved branches. The higher-truncation roots remain nulls with singular-value ratios from 4.5×10^{-17} to 8.0×10^{-13} ; no discontinuous branch connection is drawn in the main figure. The strict endpoint is separately fixed by Eq. (S15).

ISOLATED-CAVITY AND NONPERIODIC-DIMER MODES

To expose the physical origin without invoking the periodic lattice, we solve three auxiliary outgoing eigenproblems on an infinite rigid baffle: the large groove alone, the small groove alone, and the two grooves at their final separation. These calculations retain the aperture radiation integral and the full transverse cosine basis; no end correction is fitted.

The large groove supports an odd transverse ($q = 1$ -dominated) pole close to the Rayleigh-BIC frequency. The small groove supplies a much broader $q = 0$ -dominated response. Placing the grooves together hybridizes these families and reduces their radiative linewidth, but the nonperiodic dimer is still leaky. The results are summarized in Table S3.

At the dimer aperture, the surface-pressure fractions are 0.646 and 0.354 for the large and small grooves, respectively. The small groove therefore cannot be judged by its volume fraction: its broad pole contributes a substantial surface amplitude and phase shift. Periodic repetition then supplies the Bloch-dependent self- and inter-cell coupling required to move the dimer pole onto the real Rayleigh line.

TWO-COORDINATE SOURCE REDUCTION

The mechanism plot in Fig. 3 of the Letter is a reduction of the full modal-matching eigenvector, not an independently fitted 2×2 temporal coupled-mode Hamiltonian. We group the dominant aperture coefficients into large- and small-groove coordinates. For exterior order n , their contributions to the velocity-continuity row are

$$J_n = j_{nL} + j_{nS}, \quad j_{nl} = i \sum_{q \in \ell} D_{n,\ell q}(V_d)_{\ell q, \ell q} C_{\ell q}. \quad (\text{S17})$$

TABLE S2. Representative continuation points around the positive- κ Rayleigh BIC.

κ	θ (deg)	$\text{Re } f_p$ (kHz)	f_{RA} (kHz)	Γ (Hz)	$ A_0 $	$ A_{-1} $
0.1079819235	6.96974	179.974687	180.361263	2.35×10^{-3}	1.39×10^{-4}	8.25×10^{-5}
0.1099801053	7.09838	179.999751	180.003573	2.36×10^{-7}	1.39×10^{-6}	7.42×10^{-8}
0.1100001053	7.09966	180.000000	180.000000	0	2.58×10^{-16}	8.09×10^{-17}
0.1100201053	7.10095	180.000249	179.996428	3.31×10^{-7}	1.39×10^{-6}	7.35×10^{-8}
0.1120182871	7.22960	180.025024	179.640250	1.19×10^{-2}	1.41×10^{-4}	7.35×10^{-5}

TABLE S3. Outgoing poles of the final cavities on an infinite rigid baffle. The sign of $\text{Im } f$ follows the $e^{+i\omega t}$ convention.

system	dominant family	$\text{Re } f$ (kHz)	$\text{Im } f$ (kHz)	Q	large-groove fraction	small-groove fraction
large groove	odd $q = 1$	180.26568	2.09958	42.93	1	0
small groove	$q = 0$	186.97086	47.66565	1.96	0	1
nonperiodic dimer	hybrid	180.21295	1.21504	74.16	0.861	0.139

Here $D_{n,\ell q}$ is the aperture-overlap matrix in the lower block of Eq. (??). For a finite-flux order,

$$A_n = -\frac{k_0}{k_{y,n}} J_n. \quad (\text{S18})$$

Consequently $J_0 = 0$ is equivalent to $A_0 = 0$. At the Rayleigh endpoint, however, $k_{y,-1} = 0$: $J_{-1} = 0$ enforces the normal-velocity condition but does not alone determine the grazing pressure. The independent full-operator calculation gives $A_{-1} = 0$ as reported in Eqs. (S13) and (S15).

The exact endpoint eigenvector gives

$$|j_{0L}|/|j_{0S}| = 1, \quad \arg(j_{0L}/j_{0S}) = \pi, \quad (\text{S19})$$

$$|j_{-1,L}|/|j_{-1,S}| = 1, \quad \arg(j_{-1,L}/j_{-1,S}) = \pi. \quad (\text{S20})$$

The normalized complex closure errors are 2.58×10^{-16} for $n = 0$ and 8.09×10^{-17} for $n = -1$. Equation (S20) is the quantitative content of Fig. 3(c,d) in the Letter.

For one internal coordinate, both source rows can vanish only if its coupling to each row vanishes separately. With two coordinates, a relative complex weight can cancel one row, while geometry and Bloch phase can align the cancellation conditions of both rows. This is a degree-of-freedom interpretation of the present structure, not a universal theorem that every Rayleigh BIC must contain two geometrically distinct cavities.

BOUNDED SINGLE-GROOVE COMPARISON

We also searched a declared family of one large rectangular groove per period, retaining transverse modes. At the exact $n = -1$ Rayleigh relation, the search covered

$$\begin{aligned} 0.04 \leq \kappa \leq 0.25, \quad 0.04 \leq d/a \leq 2.0, \\ 0.50 \leq w/a \leq 0.95. \end{aligned} \quad (\text{S21})$$

A deterministic 1800-point survey followed by 14 bounded local refinements gave a best high-truncation residual

$$(\sigma_{\min}/\sigma_{\max})_{\text{one groove}} = 0.124588 \quad (\text{S22})$$

at $(N, K) = (221, 44)$, compared with 4.91×10^{-16} for the two-groove state. The best one-groove vector had a transverse-mode fraction of 0.961, so its failure is not caused by a single-mode cavity approximation. This result shows the practical role of the second cavity within the searched rectangular family; it is not a no-go theorem for arbitrary compound or internally structured single resonators.

FIELD LOCALIZATION AND DIMENSIONAL SIMILARITY

The eigenfields in Fig. 2(e) were reconstructed from the homogeneous coefficients on five periods and decomposed by Floquet order. All states use the same normalization. At the Rayleigh BIC, the $n = 0$ and $n = -1$ fields are identically absent, and $n = +1$ decays exponentially. Below the opening, only $n = 0$ is propagating; above it, both $n = 0$ and $n = -1$ propagate. This comparison makes the channel topology visible without assigning power to an evanescent order.

The ideal rigid-fluid equations are dimensionally scale invariant. If all lengths are multiplied by s while the normalized geometry is fixed,

$$f_{\text{BIC}}(c, s) = \frac{\Omega_{\text{BIC}} c}{s a_0} = 180 \text{ kHz} \left(\frac{c}{1500 \text{ m s}^{-1}} \right) \frac{1}{s}. \quad (\text{S23})$$

Thus, with frequency retuning, $c = 1400\text{--}1600 \text{ m s}^{-1}$ maps to $f_{\text{BIC}} = 168\text{--}192 \text{ kHz}$ for the same dimensions. This is an exact similarity relation, not robustness of a

fixed-frequency device: changing c while holding f , geometry, and angle fixed detunes both the Rayleigh condition and the radiation zero.

SCATTERING OBSERVABLES AND EXPERIMENTAL DATA GATE

The driven maps in Fig. 4 were evaluated with $(N, K) = (345, 43)$ on 91 angles and 57 adaptive detunings per angle. The maximum lossless energy-closure error was 5.21×10^{-5} . The exact BIC row is left dark because a strict homogeneous BIC cannot be excited by an ideal external plane wave. The representative field in Fig. 4(e) is a nearby driven state at $\theta = 7.74434^\circ$ and $f = 180.12125$ kHz, for which the two propagating efficiencies are $\eta_0 = 0.58713$ and $\eta_{-1} = 0.41287$.

The complex field at a scan height y_s can be projected as

$$p_{\text{sc}}(x, y_s) = \sum_n \tilde{A}_n(y_s) e^{-ik_{x,n}x}, \quad (\text{S24})$$

$$A_n = \tilde{A}_n(y_s) e^{+ik_{y,n}y_s}.$$

Equation (S24) separates the directly measured grazing pressure amplitude from its zero normal power.

In the current repository, the open symbols and sampled field in Fig. 4 are exact theory subsamples, not measurements. Before the experimental switch is enabled, they must be replaced by measured complex-pressure data and the following metadata must be inserted: number of periods, source aperture, angular calibration, hy-

drophone spacing, water temperature and sound speed, complex reference calibration, dimensional metrology, repeated scan count, and uncertainty definition. No experimental agreement is claimed in the present compile.

DATA AND CODE PROVENANCE

The final geometry and endpoint operator are stored in `Ni2019_MATLAB/results/StrictRayleighBIC_180kHz_7p10deg_final.mat`. The isolated-cavity, nonperiodic-dimer, outgoing-pole, channel-balance, eigenfield, and scattering data are stored under `Ni2019_MATLAB/results/`. Principal generation scripts are

- `scan_Ni2019_all_poles_positive_180k.m`;
- `export_Ni2019_physical_pole_spectrum_180k.m`;
- `run_target_branch_physical_bz_180k.m`;
- `export_fig2_three_state_fields_180k.m`;
- `run_Ni2019_isolated_two_groove_eigenmodes.m`;
- `run_fig3_radiation_contributions_180k.m`;
- `run_fig4_experimental_observables_180k.m`.

Figure source data are CSV/MAT files; vector PDF/SVG outputs and 600-dpi previews are retained with the plotting scripts.